

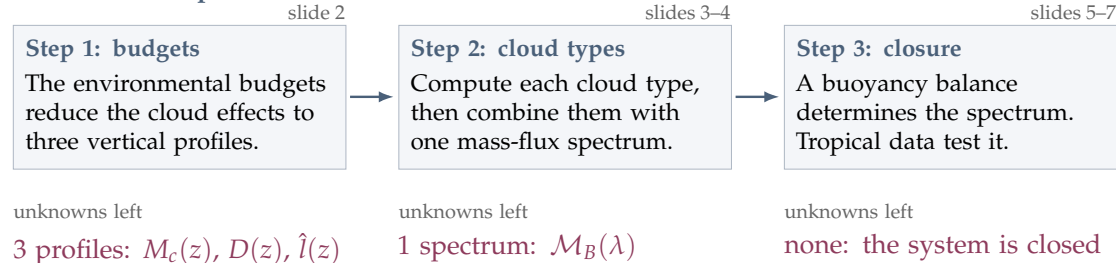
Arakawa–Schubert (1974): goal of the paper

1

Large-scale models need the combined heating and moistening effects of unresolved cumulus clouds.

Goal Compute these cloud effects from the large-scale atmosphere itself.

How: three steps



The environment sets the cloud activity, which in turn changes the environment

Arakawa–Schubert (1974): goal of the paper

Interaction of a Cumulus Cloud Ensemble with the Large-Scale Environment, Part I, Arakawa and Schubert (1974), J. Atmos. Sci. 31, 674–701. A large-scale disturbance contains many cumulus clouds whose time and space scales are much smaller than its own. Because of this scale separation, the paper aims to predict the large-scale fields by describing only the collective influence of the clouds, not each cloud. The paper calls this the goal of cumulus parameterization. The result is meant for prognostic models of the large-scale atmosphere, such as the UCLA general circulation model.

The reduction has three steps. First, budgets for the environment show that the clouds change its dry static energy $\bar{s}$ and water vapour $\bar{q}$ only through three profiles: the total cloud mass flux $M_c(z)$, the total detrainment $D(z)$ and the liquid water $\hat{l}(z)$ in the detraining air. In the subcloud mixed layer they act only on its depth. Second, a spectral ensemble of entraining plumes computes all three profiles from the environment and a single unknown function, the mass flux $\mathcal{M}_B(\lambda)$ of each cloud type at the top of the mixed layer. Third, the quasi-equilibrium of the cloud work function gives an integral equation for $\mathcal{M}_B(\lambda)$, and the system is closed. Slide 7 shows the observational test of quasi-equilibrium.

The theory describes a two-way interaction: the clouds modify the environment (step 1), and the environment controls the clouds (steps 2 and 3). Earlier parameterizations were largely empirical. Ooyama (1971) had reduced the problem to a cloud-base dispatcher function but left its determination open, so his parameterization was not closed. The bottom line states this two-way interaction. Source: Abstract, Sections 1 and 8.

Step 1: clouds enter through three profiles

2

Budgets of the air between clouds: the model knows every term except M_c , D and $\hat{l}$.

$$\rho \partial_t \bar{s} = D(\hat{s} - \bar{s} - L\hat{l}) + M_c \partial_z \bar{s} + \text{LS}_s$$

$$\rho \partial_t \bar{q} = D(\hat{q}^* + \hat{l} - \bar{q}) + M_c \partial_z \bar{q} + \text{LS}_q$$

Symbols

$s = c_p T + gz$, q dry static energy; vapour mixing ratio

$\bar{s}$, $\bar{q}$ large-scale means, i.e. the environment

$M_c(z)$ total upward cloud mass flux per unit area

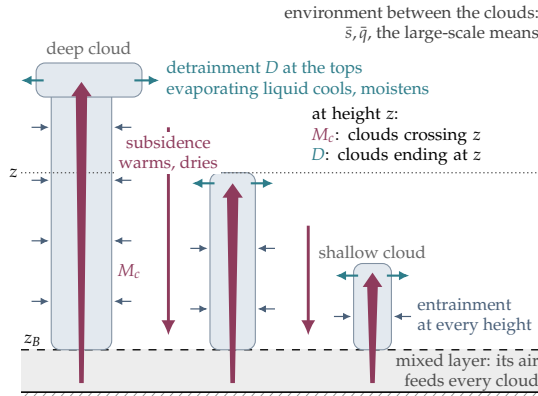
$D(z)$ mass detrained per unit volume and time

$\hat{s}$, $\hat{q}^*$ detrained-air values (star: saturation)

$\hat{l}(z)$ liquid water in the detrained air

L , ρ latent heat of vaporization; air density

LS large-scale advection (+ radiation for s)



Unknowns left: three profiles, $M_c(z)$, $D(z)$ and $\hat{l}(z)$

Step 1: clouds enter through three profiles

These are the equations the large-scale model must integrate. All terms are known to it except the cloud terms. Cloud i covers the area fraction σ_i and carries the mass flux $\rho \sigma_i w_i$. The sum M_c drives compensating environmental subsidence. In a stably stratified atmosphere ($\partial_z \bar{s} > 0$) this warms, and since $\partial_z \bar{q} < 0$ it dries. Detrainment supplies cloud air and liquid water, whose evaporation cools ($-L\hat{l}$) and moistens. The latent heat released inside the clouds does not warm the environment directly: it keeps the clouds buoyant and so maintains M_c and the subsidence warming.

The variables are the dry static energy $s = c_p T + gz$, conserved by a dry parcel moving adiabatically, and the water-vapour mixing ratio q . Hats denote detraining-air values, the star denotes saturation, and L is latent heat. $\hat{s}$ and $\hat{q}^*$ are not further unknowns: Eqs. (72)–(73) express them through $\bar{s}$, $\bar{q}$, $\bar{q}^*$ and $\hat{l}$. For legibility the slide groups the large-scale terms as $\text{LS}_s = -\rho \bar{\mathbf{v}} \cdot \nabla \bar{s} - \rho \bar{w} \partial_z \bar{s} + \bar{Q}_R$ and $\text{LS}_q = -\rho \bar{\mathbf{v}} \cdot \nabla \bar{q} - \rho \bar{w} \partial_z \bar{q}$. $\bar{Q}_R$ is the radiative heating, including that inside the clouds, which the paper sets aside.

In the subcloud mixed layer the clouds do not act on temperature and moisture directly. They drain mass through its top, and this cumulus-induced subsidence makes it shallower (Eq. 126). The equations assume a small active cloud fraction and a horizontally uniform environment. The figure redraws the paper's conceptual Fig. 1; it is a schematic, not a measured cloud map. Read it with the equations: the wine arrows are the mass flux M_c rising in the clouds and the subsidence it forces between them, the teal arrows are the detrainment D at the cloud tops. The updraft arrows widen upward because entrainment adds mass at every height. At the dotted level z , $M_c(z)$ sums the deep and medium clouds that cross it, and $D(z)$ comes from the medium cloud whose top is there: this is why the cloud terms are profiles in z . Section 4 opens with the conclusion on the slide: the parameterization problem is reduced to finding $M_c(z)$, $D(z)$ and $\hat{l}(z)$. Source: Sections 2–4, Figure 1, Equations (7), (12), (72)–(75) and (126).

Step 2: the structure of one cloud type

3

Entrainment dilutes a rising plume. The more it entrains, the lower its top.

$$\partial_z \eta = \lambda \eta$$

mass grows: $\eta = e^{\lambda(z-z_B)}$

$$\partial_z h_c = \lambda(\bar{h} - h_c)$$

h_c is pulled toward $\bar{h}$ over a length $1/\lambda$

$$h_c(z_D) = \bar{h}^*(z_D)$$

top $z_D(\lambda)$: buoyancy vanishes

All types start at z_B with mixed-layer air: $\eta = 1$, $h_c = h_M$.

Symbols

λ fractional entrainment rate: the cloud-type label

η mass flux normalized at the mixed-layer top z_B

$h = s + Lq$ moist static energy; h_M : its mixed-layer value

$\bar{h}$, $\bar{h}^*$ environment h ; the h it would have if saturated

$h_c > \bar{h}^*$ plume warmer than the environment, so buoyant

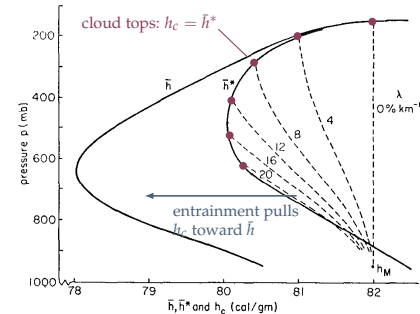


Fig. 6, tropical sounding; up is higher. Dashed: plume h_c for $\lambda = 0$ –20% km^{-1} , all from h_M . Dots (added): cloud tops.

Given the environment and λ , the plume model fixes η , h_c and the cloud top z_D

Step 2: the structure of one cloud type

Cloud types are labelled by the fractional entrainment rate λ , constant with height. Each type starts at the mixed-layer top z_B with mixed-layer properties, $h_c(z_B, \lambda) = h_M$ (Eq. 131). The first equation gives $\eta = e^{\lambda(z-z_B)}$ up to the top: entrainment adds mass as the plume rises. The second is the moist-static-energy budget of an entraining plume (Eqs. 83–89), with $h = s + Lq$: entrained air pulls h_c toward the environmental $\bar{h}$. Its integral form is $\eta h_c = h_M + \lambda \int_{z_B}^z \eta \bar{h} dz'$ (Eq. 97). Dividing by η gives $h_c(z) = e^{-\lambda(z-z_B)} h_M + \lambda \int_{z_B}^z e^{-\lambda(z-z')} \bar{h}(z') dz'$: a weighted mean of the mixed-layer value and of the environment below z , with the nearest levels weighted most. The plume forgets h_M over a length $1/\lambda$, 5 km for $\lambda = 20\% \text{ km}^{-1}$; for $\lambda = 0$ it keeps h_M . It never reaches $\bar{h}$: $\bar{h}$ changes with height, and since $\bar{h} < \bar{h}^*$ the plume stops first, where it meets $\bar{h}^*$.

Why $h_c > \bar{h}^*$ means buoyant: the cloud is saturated, and Eq. (98) gives $s_c - \bar{s} = (h_c - \bar{h}^*) / (1 + \gamma)$ with $\gamma = (L/c_p)(\partial q^* / \partial T)_p > 0$. At the same height $s_c - \bar{s} = c_p(T_c - \bar{T})$, so the plume is warmer than its surroundings exactly when h_c lies right of $\bar{h}^*$, and the top is where they meet. This neglects the vapour and liquid-water corrections, written out on slide 5, and the small difference between $\hat{h}^*$ and $\bar{h}^*$.

Reading Figure 6 (Jordan's mean hurricane-season sounding). The vertical axis is pressure, so up is higher. Solid curves are the environment: $\bar{h}$ has its minimum near 650 mb, the dry middle troposphere; $\bar{h}^*$ is the value it would have if saturated. Every plume starts at 950 mb with $h_M = 82 \text{ cal g}^{-1}$. The line $\lambda = 0$ is undiluted and meets $\bar{h}^*$ only near the tropopause. As λ grows, the plume is pulled toward the low $\bar{h}$, crosses $\bar{h}^*$ earlier and has a lower top, around 600 mb for $\lambda = 20\% \text{ km}^{-1}$ ($= 0.2 \text{ km}^{-1}$). The horizontal gap to $\bar{h}^*$ is proportional to the temperature excess. The wine dots, added on the slide, mark these crossing points; the arrow marks the leftward pull of entrainment. The paper's Fig. 8 collects the crossing points as $p_D(\lambda)$. Simpson's tower model gives $\lambda = 2\alpha/R$ (Eq. 94): large λ means a narrow cloud. Source: Sections 4–5, Equations (83)–(98), (129)–(131), Figures 6–8.

Step 2: one spectrum gives the three profiles

4

The plume model gives each type's structure; the spectrum $\mathcal{M}_B(\lambda)$ weights the types.

Clouds still rising at height z

$$M_c(z) = \int_0^{\lambda_D(z)} \mathcal{M}_B(\lambda) \eta(z, \lambda) d\lambda$$

Clouds ending at height z

$$D(z) = -\mathcal{M}_B(\lambda_D) \eta(z, \lambda_D) \frac{d\lambda_D}{dz}$$

$$\hat{l}(z) = l(z, \lambda_D)$$

Symbols

$\mathcal{M}_B(\lambda) d\lambda$ mass flux at z_B of the types in $(\lambda, \lambda + d\lambda)$

$\eta(z, \lambda)$ normalized mass flux (slide 3)

$\lambda_D(z)$ type whose cloud top is at z

$l(z, \lambda)$ liquid water inside type λ (below)

Where l comes from

Plume water budget (Eq. 85): vapour in by entrainment, rain r out.

$$\partial_z [\eta (q_c + l)] = \lambda \eta \bar{q} - \eta r$$

Cloud vapour q_c is saturated (Eq. 90), so the rest is l . Crude rain: $r = C_0 l$ (App. B).

Only the mass-flux spectrum $\mathcal{M}_B(\lambda)$ remains to be determined

Step 2: one spectrum gives the three profiles

Summing the cloud types gives the three profiles needed in the environmental budgets. The spectrum is a mass-flux density over entrainment rate: $\mathcal{M}_B(\lambda) d\lambda$ is the mass flux at the mixed-layer top of all clouds with entrainment rate between λ and $\lambda + d\lambda$ (the paper's sum over $\lambda_i \in (\lambda, \lambda + d\lambda)$ evaluated at z_B , Eqs. 78 and 81). Multiplication by $\eta(z, \lambda)$ gives its mass flux at height z .

$M_c(z)$ integrates the types still rising at that level, those with $\lambda < \lambda_D(z)$ (Eq. 77). Detrainment $D(z)$ comes from the types whose top lies at z (Eq. 79). The derivative converts a cloud-type interval to a height interval, and the minus sign gives positive detrainment because λ_D decreases with height. Each plume detrains all its mass in a thin layer at its top. The detrained liquid water is that of the type ending there, $\hat{l}(z) = l(z, \lambda_D)$ (Eq. 91). The liquid water comes from the total-water budget of the plume, $\partial_z [\eta (q_c + l)] = \lambda \eta \bar{q} - \eta r$ (Eq. 85, using $\partial_z \eta = \lambda \eta$): entrainment brings in environmental vapour, rain r removes water. Above cloud base the cloud vapour q_c is the saturation value, fixed by h_c (Eq. 90), so what remains of the total water is liquid. Appendix B closes the budget with the crude rule $r = C_0 l$, a constant fraction C_0 rained out per unit height, which gives $\eta l = \int_{z_B}^z e^{-C_0(z-z')} \{-\partial_{z'}(\eta q_c) + \lambda \eta \bar{q}\} dz'$ (Eq. B7): water condensed at each level, depleted exponentially by rain on the way up.

For a given environment and these model assumptions, the unknown cloud contribution has therefore reduced from three height profiles to one mass-flux spectrum. This spectrum does not separately prescribe cloud number or lifetime. The next two slides introduce the energy measure and the balance used to determine it. Source: Sections 4–5, Equations (77)–(91), (129)–(131), Appendix B (B7).

Step 3: the cloud work function

5

The work function $A(\lambda)$ adds up buoyancy over the cloud depth, weighted by mass flux. For each cloud type, the environment determines A .

$$A(\lambda) = \int_{z_B}^{z_D(\lambda)} \underbrace{\eta(z, \lambda)}_{\text{weight}} \underbrace{\frac{g}{c_p \bar{T}(z)} [s_{vc}(z, \lambda) - \bar{s}_v(z)]}_{\text{buoyancy}} dz$$

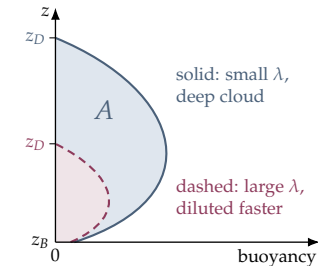
Symbols

$s_v = s + c_p \bar{T} (0.608 q - l)$ virtual static energy (Eq. 58)

T, q, l temperature, vapour, liquid water; c : cloud, bar: environment

g, c_p gravity; specific heat of air at constant pressure

η mass flux normalized at the mixed-layer top z_B



Schematic. A : area under the buoyancy curve, weighted by η .

Why “work”: kinetic energy
 $dK/dt = A \mathcal{M}_B - \mathcal{D}$

Positive A allows buoyant energy production, but does not set the cloud activity

Step 3: the cloud work function

The cloud work function is the buoyancy integral weighted by normalized mass flux. It measures kinetic-energy generation per unit mass flux at the reference level, as the kinetic-energy budget reads $dK/dt = A \mathcal{M}_B - \mathcal{D}$, with $\mathcal{K}$ the kinetic energy and $\mathcal{D}$ dissipation. It is not the kinetic energy already stored in the clouds. Its units are energy per mass, for example J kg^{-1} .

The integral is Eq. (133). The buoyancy term uses the virtual static energy $s_v = s + c_p \bar{T}(\delta q - l)$ with $\delta = 0.608$ (Eq. 58). Water vapour lightens the air and liquid water adds weight. The subscript c denotes the cloud and the bar denotes the environment. The normalized mass flux η weights the contribution of each height. The paper’s later algebra (Eqs. 134–139) temporarily drops the virtual corrections and writes buoyancy through $h_c - \bar{h}^*$, the gap seen in Fig. 6 on slide 3.

The right panel is a schematic, not data. It plots the buoyancy against height for two cloud types: the shaded area is what A adds up, with each height weighted by η . A larger λ dilutes the plume faster, so its buoyancy is weaker and vanishes lower, as in Fig. 6. The kinetic-energy budget of Eq. (132) explains the name: buoyancy generates kinetic energy in type λ at the rate $A \mathcal{M}_B$, so A is the kinetic energy produced per unit mass flux.

The plume model determines buoyancy and the cloud top for a given environment, including the mixed layer. Thus $A(\lambda)$ is a property of that environment for each type. Positive A permits buoyant energy production, but production also requires cloud activity, $\mathcal{M}_B > 0$, and triggering. For a non-entraining cloud, $\lambda = 0$, the integral is close to modern CAPE. Entraining clouds also depend on environmental moisture. Source: Sections 3 and 6, Equations (58)–(59), (132)–(139).

Step 3: the quasi-equilibrium closure

6

The rate of change of A is the sum of the large-scale forcing and the cloud response.

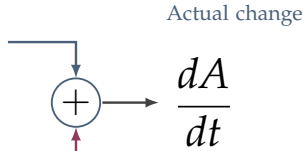
$$\frac{dA(\lambda)}{dt} = \int_0^{\lambda_{\max}} K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda' + F(\lambda)$$

Forcing $F(\lambda)$

Large-scale processes

Cloud response

All types λ' , through K



Kernel $K(\lambda, \lambda')$

Change of $A(\lambda)$ per unit time due to unit cloud-base mass flux of type λ' , acting through the environment. Usually $K < 0$: clouds use up A .

K_V mostly < 0 : subsidence driven by λ' warms the air, so λ is less buoyant

$K_D \geq 0$: shallower clouds ($\lambda' > \lambda$) detrain, cooling and moistening the air of deeper ones

K_M mass drawn up at z_B changes the mixed-layer depth

Quasi-equilibrium: cloud and forcing terms nearly cancel, giving $dA/dt \simeq 0$

Step 3: the quasi-equilibrium closure

The tendency equation separates the cloud response, $\int_0^{\lambda_{\max}} K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda'$, from the large-scale forcing F . The prime labels the cloud type producing the response, while unprimed λ labels the work function affected. F includes large-scale ascent and advection, radiation and surface fluxes through both the cloud layer and mixed layer.

The diagram reads Eq. (142) as a sum of two signed rates, forcing and cloud response, giving the actual tendency dA/dt . The cloud term carries its sign through K . It is a balance of tendencies, not a sequence of causal steps.

The kernel (Eq. 141): $K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda'$ is the rate at which type- λ' clouds change $A(\lambda)$ by modifying the environment. Each type changes the air that all other types rise through, so each work function feels all types. The paper splits it into three parts (Eq. 144). The vertical mass-flux kernel K_V is dominated by the subsidence that type λ' forces between the clouds: it warms the environment, so type λ is less buoyant and $A(\lambda)$ falls. Neglecting smaller effects, K_V is symmetric (Fig. 11). The detrainment kernel K_D is zero for $\lambda' < \lambda$, because deeper clouds detrain above the top of type λ , and nearly always positive for $\lambda' > \lambda$: shallower clouds detrain saturated air and liquid water whose evaporation cools and moistens the layer that deeper clouds cross, raising their buoyancy (Fig. 12). The mixed-layer kernel K_M carries the effect of the cloud-base mass flux on the mixed-layer depth z_B . The paper calls K typically negative: the net effect of clouds is to use up A .

With the timescale separation of slide 7, dA/dt is small compared with the cloud and forcing terms. Dropping it gives the integral equation for the unknown spectrum. The equality applies to active types with $\mathcal{M}_B > 0$. Absent types satisfy the paper's inequality, and the full spectrum is nonnegative (Eq. 158). Temperature and moisture can still evolve.

Cloud number, lifetime and single-cloud mass transport are not needed separately: only their combination $\mathcal{M}_B$ enters (Eq. 145). The closure, a Fredholm integral equation of the first kind, is the main result of Part I; its solution is deferred to Part II.

Source: Sections 6–7, Equations (141)–(145), (146)–(150), (158), Figures 11–12, and Section 8.

Quasi-equilibrium: timescales and observations

dA/dt : change of A per unit time. Observed residual: $|dA/dt| \ll |F_C|$.

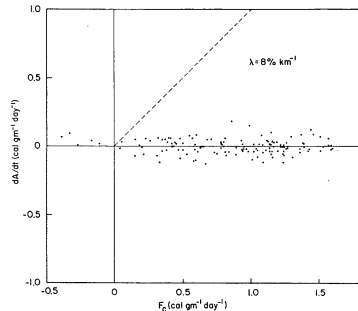


Fig. 13. F_C : cloud-layer forcing. Dashed line: $dA/dt = F_C$ without cloud response.

τ_{adj} : cloud adjustment time

Defined in words (§7): time for the clouds alone, with no forcing, to bring A to neutral ($A = 0$). Estimated from typical A and cloud term (Eqs. 152–154): $\tau_{\text{adj}} \sim 10^3$ – 10^4 s, 15 min–3 h.

τ_{LS} : forcing time

Defined in words: time over which F changes significantly. Stated as typical, not derived: $\tau_{\text{LS}} \gtrsim 10^5$ s, a day or more.

Why the net change is small (Eqs. 146–148)

cloud term	$ [dA/dt]_C \sim A/\tau_{\text{adj}}$	clouds could remove A in τ_{adj}
net change	$ dA/dt \lesssim A/\tau_{\text{LS}}$	A follows the slow forcing
ratio	$\lesssim \tau_{\text{adj}}/\tau_{\text{LS}} \ll 1$	cloud term and F nearly cancel

$\tau_{\text{adj}} \ll \tau_{\text{LS}}$: clouds remove A far faster than A changes, so they nearly cancel F

τ_{adj} is the response time of the whole cloud ensemble: with negligible forcing, the clouds alter their environment until A reaches zero. τ_{LS} is the time over which the large-scale forcing F changes substantially. Neither is an individual cloud lifetime.

Neither time has a formal definition. The paper introduces both in words in Section 7 and uses them only as orders of magnitude. τ_{adj} is estimated, not derived: with cloud excess temperatures of a few degrees, $A \sim (10^{-1} \text{ m s}^{-2})H_c$ (Eq. 152), with H_c the cloud depth; subsidence warming gives $[dA/dt]_C \sim (10^{-4}$ – $10^{-5} \text{ m s}^{-3})H_c$ (Eq. 153); their ratio is $\tau_{\text{adj}} \sim 10^3$ – 10^4 s (Eq. 154). $\tau_{\text{LS}} \gtrsim 10^5$ s is stated as typical, without derivation.

dA/dt is the net change, cloud plus large-scale terms. The paper estimates $|[dA/dt]_C| \sim A/\tau_{\text{adj}}$ (Eq. 146), while $|dA/dt| \lesssim A/\tau_{\text{LS}}$ (Eq. 147). Thus the net tendency is smaller than the cloud term by a factor of order $\tau_{\text{adj}}/\tau_{\text{LS}} \ll 1$ (Eq. 148). The reasoning in words: left alone, the clouds would remove all of A within τ_{adj} , so the cloud term alone is of size A/τ_{adj} . But in quasi-equilibrium A follows the current forcing, which changes only over τ_{LS} , so A itself changes at the much smaller rate A/τ_{LS} . Since dA/dt is the sum of the cloud term and F , a sum much smaller than either term means the two nearly cancel. Writing $dA/dt \simeq 0$ drops this residual; A , temperature and moisture may still evolve slowly.

Figure 13 (Marshall Islands data of Yanai et al. 1973, $\lambda = 8\% \text{ km}^{-1}$; axes in $\text{cal g}^{-1} \text{ day}^{-1}$, $1 \text{ cal g}^{-1} \approx 4.2 \text{ kJ kg}^{-1}$) plots the observed dA/dt against the cloud-layer forcing F_C . The dashed line $dA/dt = F_C$ is what forcing alone would give. The observed tendency stays near zero while F_C reaches about $1.5 \text{ cal g}^{-1} \text{ day}^{-1}$; the panel for $\lambda = 16\% \text{ km}^{-1}$ agrees. This supports quasi-equilibrium in this tropical setting, as a partial test: the mixed-layer forcing F_M , part of $F = F_C + F_M$, could not be computed from conventional data. Figure 14 shows temperature and moisture evolving while A stays small. Source: Section 7, Equations (146)–(154), Figures 13–14, and Section 8.